

Automated and Intelligent generation of Massing Models and Visualisation from Design Brief

Team 7

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Full contact details of client
Full site address of client
Details of any other important contacts in the client team

Client information:

- Who are you going to bring development?
- Why are you the participants of this project?
- What are the client's specific wishes and demands?
- Does the client have any specific wishes and demands?
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Needs:

- On what basis is the calculation of fees based?
- On what basis can the calculation be estimated and based to base?
- What is the client's?
- What is the client's?

Basic Design Questions:

- Dependent on the type of project questions will be
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About the client:

- Describe your current home. What do you like and dislike about it? What is missing, and what would you like to see in the future?
- What would you like to see in the future?
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About the site:

- Why did you choose this site?
- Is there anything about this site that you are?
- Are there any views within the site that you are?

About the occupants:

- How many people will be living in this home?
- Do you foresee more people in the home?
- Are there any pets that will share the home?

Details of any other important parties in the design process

— fax, phone number, email

THE ARCHITECTURAL DESIGN BRIEF - CHECKLIST

What is an architectural design brief?

Architectural design brief is a document that outlines the project's goals, objectives, and requirements. It serves as a guide for the architect and the client, ensuring that the design process is clear and efficient. The brief typically includes information about the project's location, size, budget, and timeline, as well as the client's specific needs and preferences. It is a crucial tool for communicating the client's vision and ensuring that the architect understands the project's requirements.

The image displays four book covers. At the top right is the 'Code of Practice' (2019 Edition) for 'Vehicle Parking Provision' in development proposals, published by the Department for Transport. Below it to the left is the 'CODE ON ACCESSIBILITY IN THE BUILT ENVIRONMENT 2019' published by the Department for Levelling Up, Housing and Communities. To the right of that is the 'CODE OF PRACTICE ON ENVIRONMENTAL HEALTH' published by the National Environment Agency. The bottom left cover is partially obscured and shows a grid-like architectural plan.

3 Apply and extract relevant parameters

[illegible]

2 Architects extract info and convert it into usable data through tables



4 Generate massing models

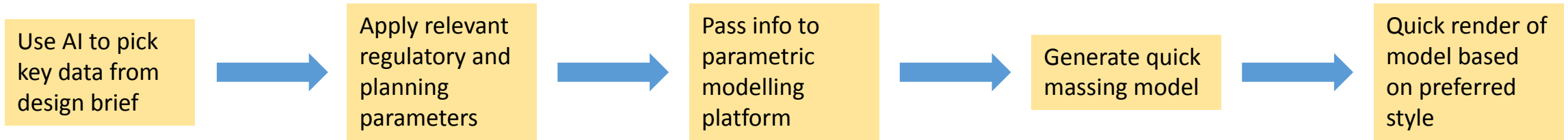
Problems

- Briefs provided in different formats
- Lengthy pieces of text
- Varying degrees of fixed and variable info and parameters

Routine and mundane task that takes time

Solutions

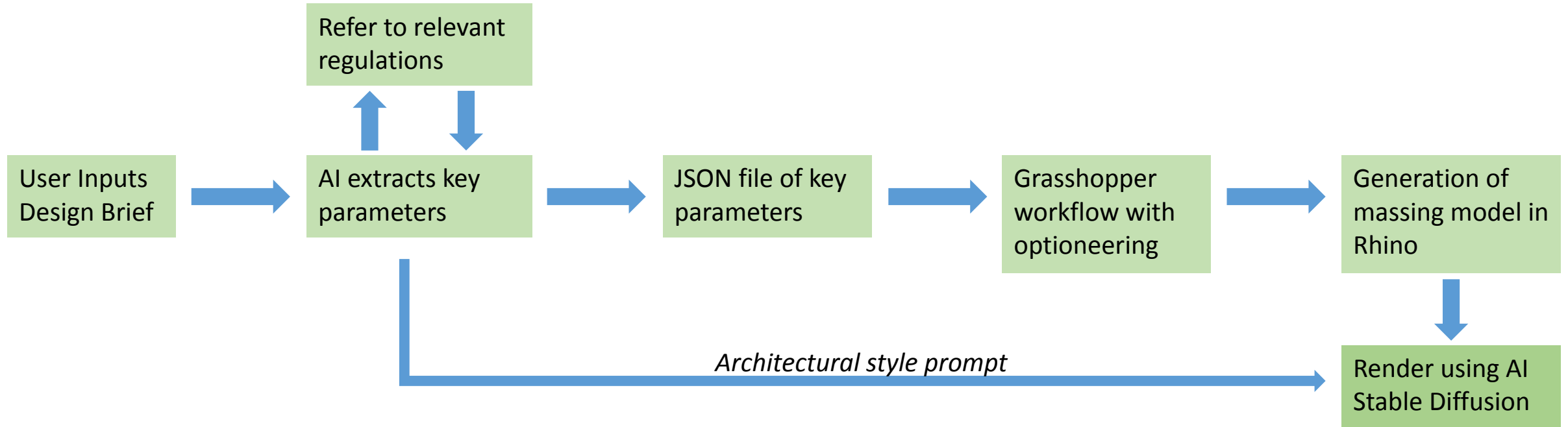
- Automation of steps 1 to 4 mentioned earlier



Objectives

- Allow architects and project members quickly get overview of project and saves time
- Assist them to determine scope of works & resource planning immediately at the onset of receiving brief
- Allow more accurate quoting of fees and high level cost estimation before project begins
- Greatly useful for feasibility studies and design competitions

Crafting the solution



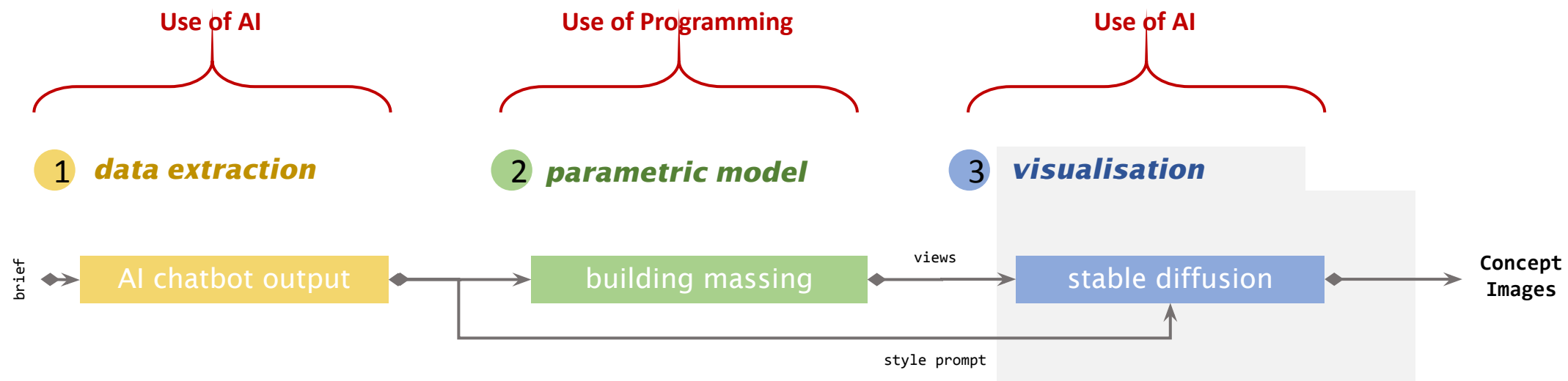
Tools used

- Botpress – A ChatGPT chatbot powered by the latest in LLMs and GPT by OpenAI
- Rhino+Grasshopper – Rhino for 3D modelling; Grasshopper for parametric visual programming
- Stable Diffusion – A deep learning, text-to-image model based on diffusion techniques

Inputs required

- Design Brief
- Site boundary in closed planar polyline

3 Step solution



1. Use AI chatbot to extract key parameters from design brief

AI Task

Task Instructions *

Extract development type, total site area, total floor area, number of blocks, number of units, max ASML, floor-to-floor height and building setback distance from the user message.

Then, convert them into JSON format.

AI Transition

Text to categorize *

{{event.preview}}

Store result in variable

Select/Create a variable

Categories *

- something positive
- something negative

> Advanced Settings

AI Generate Text

Prompt *

Extract the architectural style from the brief. Inform if this information was provided or stated in the user message, if not, give three options of architectural style that fits the location and context of the above-mentioned project in extremely simple short response without a description.

51/ ⓘ

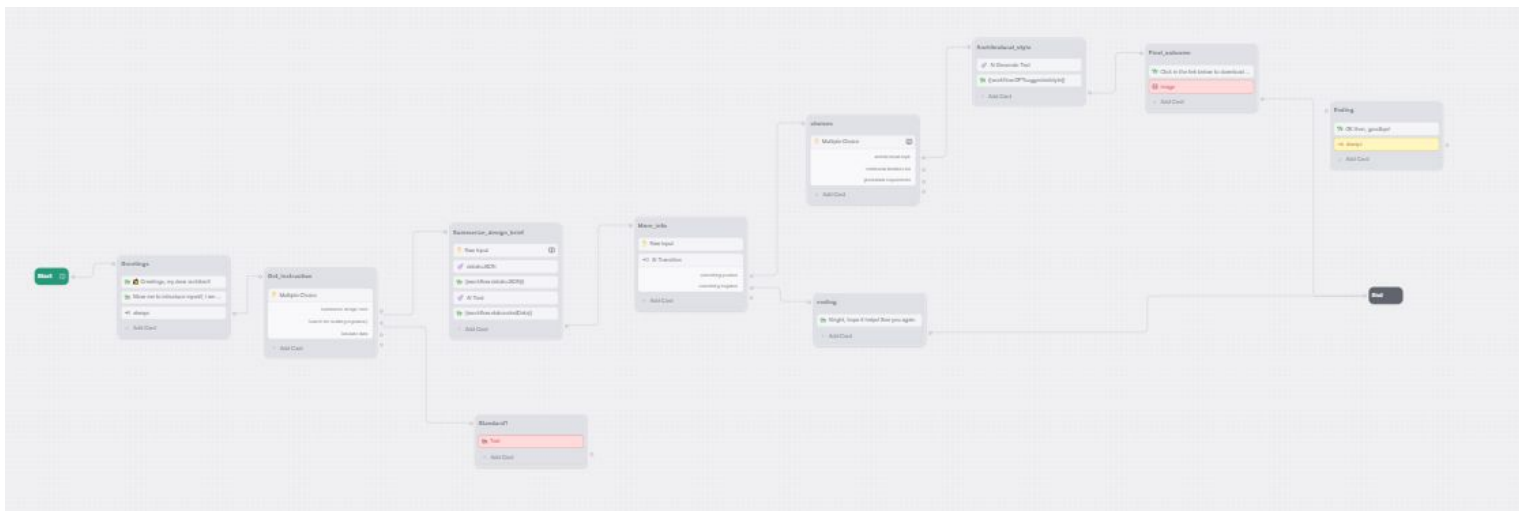
Output variable *

Tt workflow.GPTsuggestedstyle

Advanced Settings

Language Model GPT-4 Turbo

Temperature 0.20



1. Use AI chatbot to extract key parameters from design brief

A

AAA (AI Architectural Assistant)
let's chat!

Apr 14, 9:32 PM

hi

👋 Greetings, my dear architect!

Allow me to introduce myself, I am an online chatbot that helps to improve your work efficiency! .

A How can I help you ?

Summarize design brief

Search for building regulatory

Tabulate data

Chat with AAA (AI Architectural Assistant)

A

AAA (AI Architectural Assistant)
let's chat!

interior for the related water feature details such as overflow channel.

Refer to Engineer's requirement and specification.

[
{
"developmentType": "residential",
"siteArea": "500",
"floorArea": "1000",
"numberOfBlocks": 3,
"numberOfUnits": 50,
"floorToFloorHeight": 3,
"buildingSetback": 10
}
]

A Any further information you want to extract from your design brief?

Chat with AAA (AI Architectural Assistant)

A

AAA (AI Architectural Assistant)
let's chat!

"floorToFloorHeight": 3,
"buildingSetback": 10
}
]

A Any further information you want to extract from your design brief?

yes

A What do you want to know more about

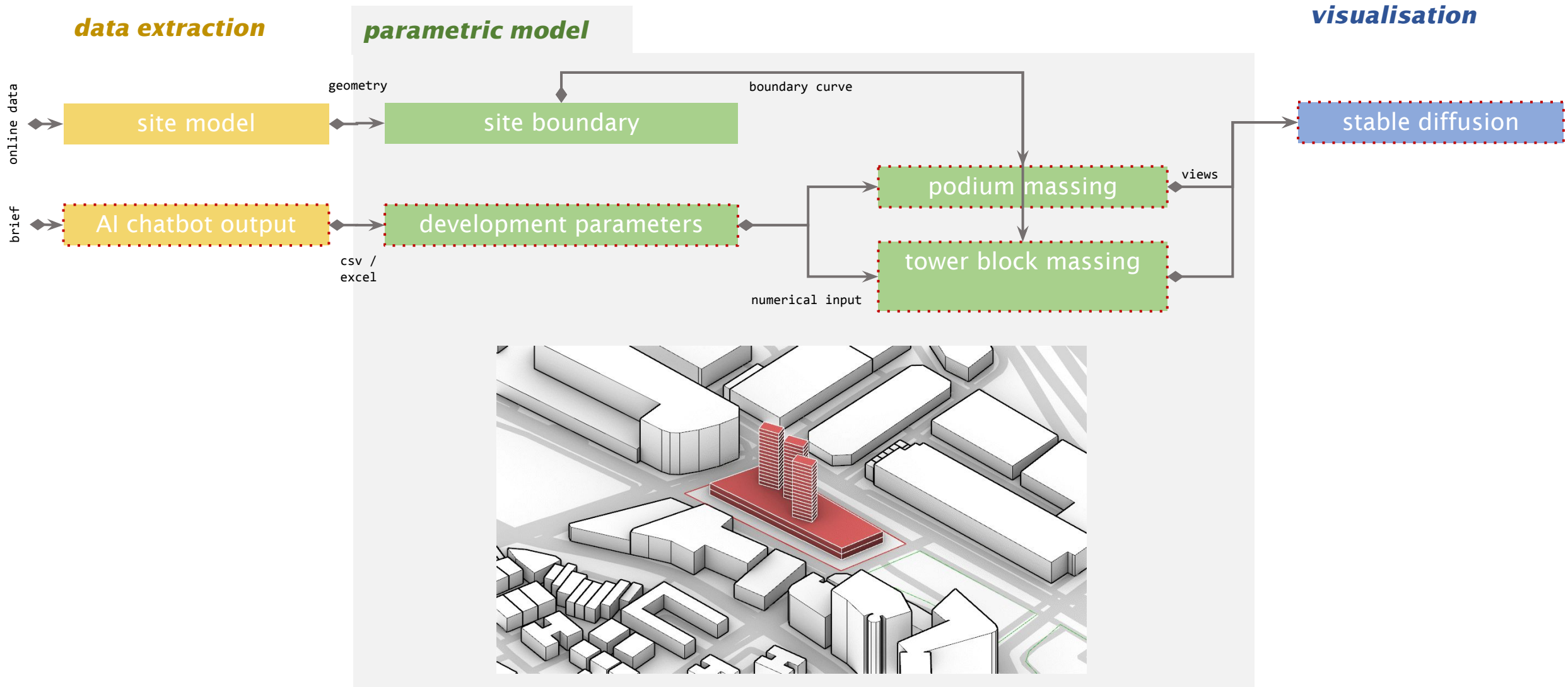
architectural style

The architectural style of the proposed residential flat development is not specified in the provided documents.

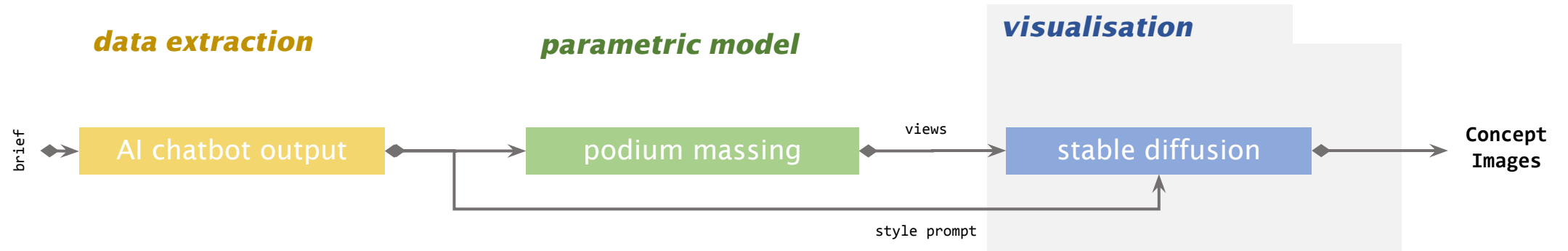
A Contemporary, Tropical, Biophilic

Chat with AAA (AI Architectural Assistant)

2. Generate massing model in Rhino+Grasshopper with extracted parameters



3. Render in Stable Diffusion based on styles extracted from design brief



From Brief to Pitch in 15 mins

data extraction

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let's chat!

Apr 14, 9:32 PM

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A How can I help you ?

Input brief

```
[
  {
    "developmentType": "residential",
    "siteArea": "500",
    "floorArea": "1000",
    "numberOfBlocks": 3,
    "numberOfUnits": 50,
    "floorToFloorHeight": 3,
    "buildingSetback": 10
  }
]
```

A Any further information you want to extract from your design brief?

yes

A What do you want to know more about

architectural style

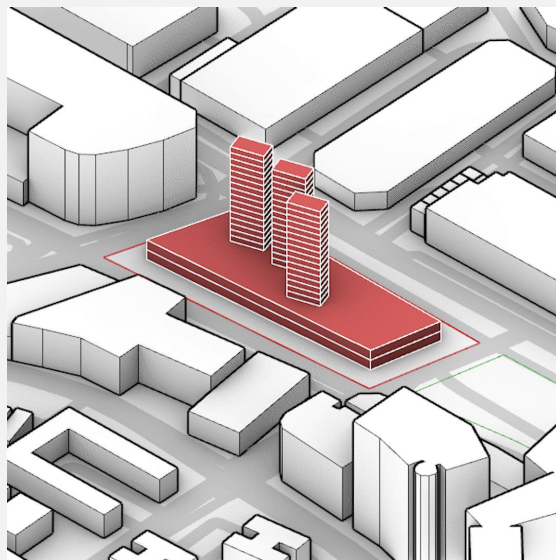
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A Contemporary, Tropical, Biophilic

Chat with AAA (AI Architectural Assistant)

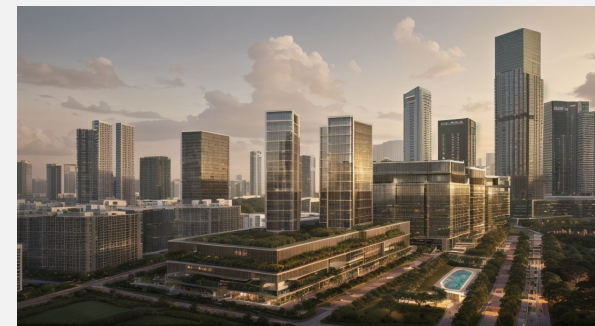
AI chatbot

parametric model



Design tool

visualisation



Stable diffusion

BRIEF

PITCH

Starting small and simple to making it more comprehensive

1

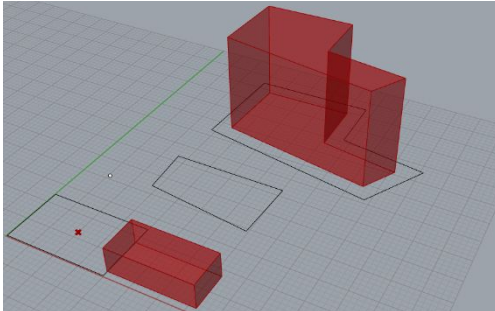
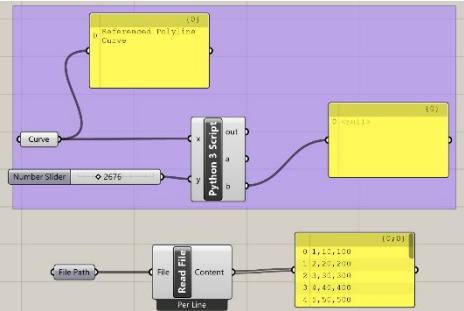
Start with simple prompts



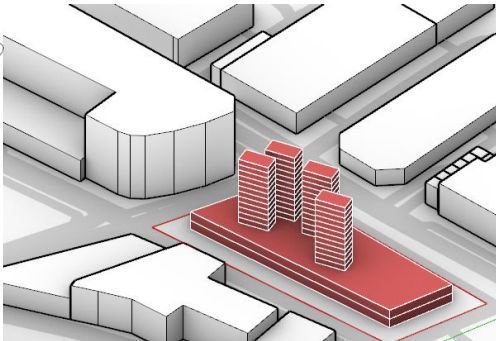
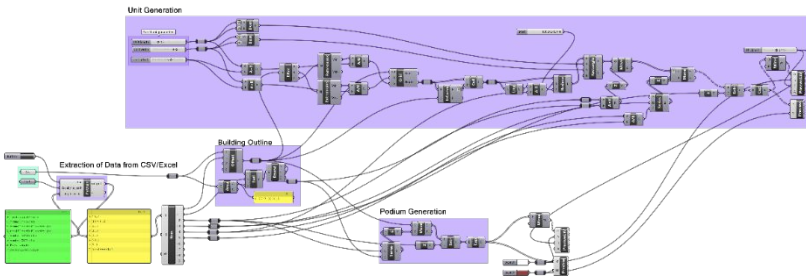
Create chatbots

2

Model generation with simple parameters

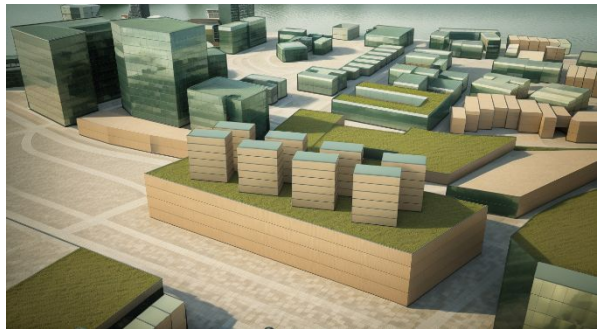


Model generation with multiple and complex parameters



3

Simple renders

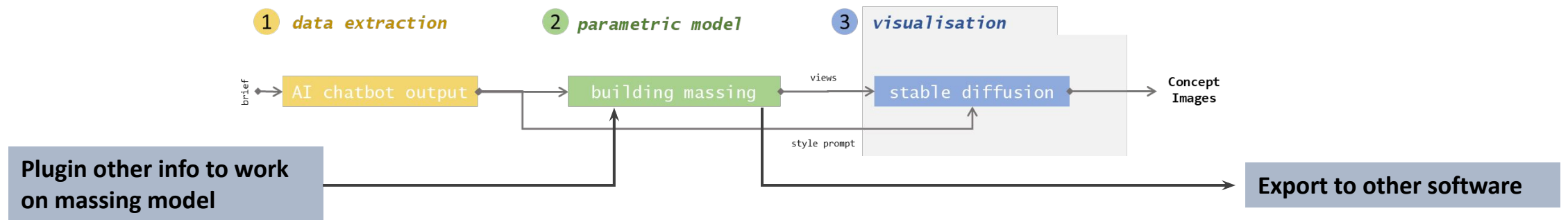


Photorealistic renders with ControlNet



Next steps to Scale up

1. **Unified interface** □ Complete solution within 1 app/ program
2. Extracting **more granular design parameters** – colours, signage etc
3. Application of more **detailed regulatory requirements** e.g. corridor widths, escape staircase determination etc
4. Expand optioneering – **AI data analysis** to select most efficient or most preferred layout based on client's requirements
5. Expand jurisdiction beyond Singapore – use AI to analyse and apply regulatory requirements for **projects overseas**
6. Picking sites directly from **Google Earth**
7. **Link final solutions to BIM softwares** such as Revit and Archicad with IFC-SG information incorporated for further design development
8. **Plugin scripts** that can work on top of generated massing model such as façade generation or parking layout generation
9. **Iterative development** with multiple stakeholders and interoperability with multiple programs to allow solution to be used from beginning to end of entire project cycles



Thank you 🥰

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